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Introduction

This Social Atlas examines the geographic distribution of voter behaviour during Singapore's 2020 General Election. While the governing People's Action Party (PAP) has been in power since 1959, this election saw the Workers' Party (WP) break the record for overall contested vote share for any opposition party with 50.49% of the votes (ELD, 2020).

This report aims to better understand the distribution of winning margins between the PAP and opposition parties, particularly WP. Furthermore, it uncovers relationships between social characteristics of electorates and their voting behaviour. First, it analyses the distribution and spatial autocorrelation of winning margins, followed by its correlation with housing type, elderly population and Chinese population, and zooms in onto WP wards. Lastly, an analysis on gerrymandering by race is also conducted.

While the PAP government claims adjusting electoral boundaries as a necessary, to ensure minority participation and ensure equitable representation (ELD, 2020), critics have raised allegations of gerrymandering — electoral manipulation to create undue advantage for the ruling party (Johnston, 2002; Ong, 2020).

In light of the upcoming general election, it is crucial for the public to critically assess geographic characteristics which may correlate with electoral outcomes, to discern whether boundary adjustments reflect demographic changes, or work to maximise the incumbent's vote share.

Subzone-level data was used as it represents the smallest administrative unit. As census and electoral data do not share boundaries, each social and constituency data have been aggregated using interpolation.

Winning Margins

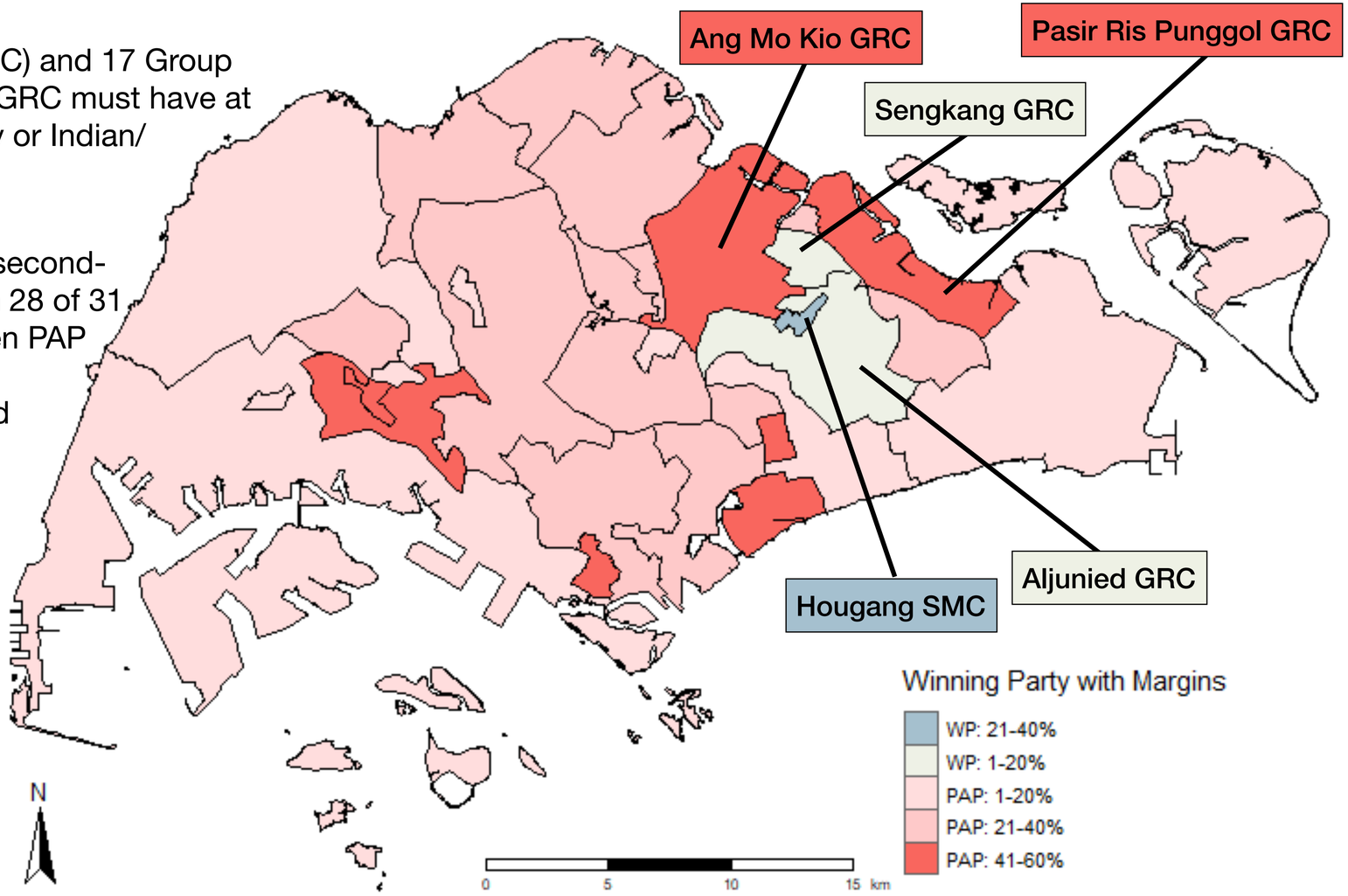
Singapore has 17 Single Member Constituencies (SMC) and 17 Group Representative Constituencies (GRC), whereby each GRC must have at least one candidate from a minority race: either Malay or Indian/ Other, depending on demographic characteristics.

While the 2020 General Election saw PAP receive its second-lowest island-wide vote share of 61.24% (ibid), it won 28 of 31 wards. The margins, defined as the difference between PAP and oppositions' vote share, varied island-wide; the vertically-centred region had larger margins compared to the far- east and west regions.

In addition to retaining Aljunied and Hougang, WP won the newly created Sengkang GRC, revealing a potential opposition stronghold in the north-east, nonetheless surrounded by heavily PAP-leaning Yuhua and Pasir Ris Punggol to the north.

This map also reveals little geographically clustering of PAP's strongest performing constituencies.

Winning Margins in Singapore's 2020 General Election (%)



Identifying Clusters

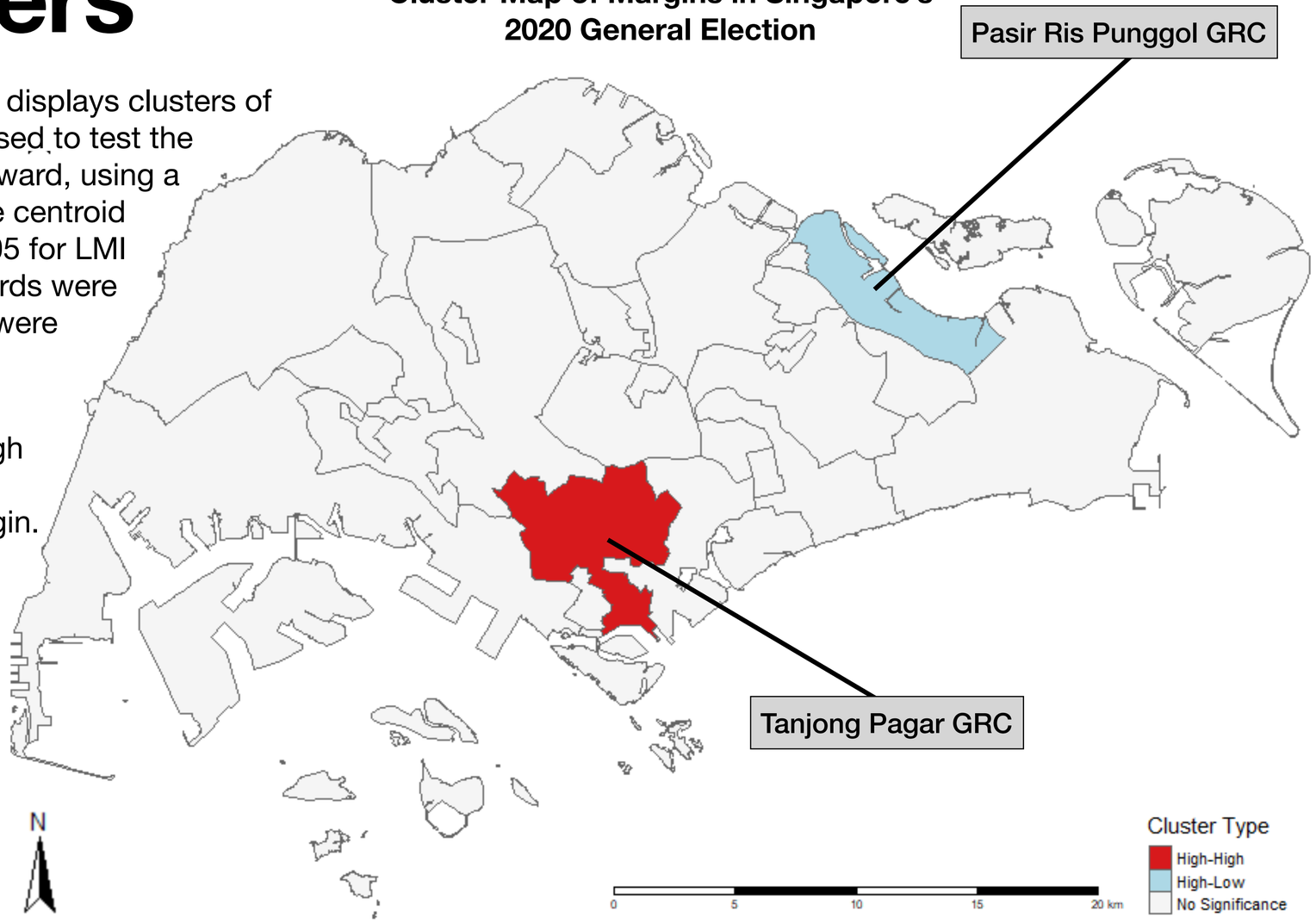
This local indicators of spatial association (LISA) map displays clusters of voting margins. A local Moran's I (LMI) statistic was used to test the randomness of spatial distribution in relation to each ward, using a fixed distance spatial weight matrix of 10km, as some centroid were on non-contiguous islands. A threshold of $p<0.05$ for LMI values were considered statistically significant. 29 wards were found to have a statistically non- significant LMI and were excluded from the LISA map.

Tanjong Pagar, a high-high cluster, saw a relatively high PAP- leaning margin and is also neighboured by constituencies with a relatively high PAP-leaning margin.

Conversely, Pasir Ris Punggol was a high-low cluster, with a relatively high PAP-leaning margin but is surrounded by constituencies with a relatively low (or negative) PAP-leaning margin, agreeing with the data in the previous map.

Due to constituencies being so large, the analysis is subject to the modifiable areal unit problem, hence smaller scale analysis is needed for greater precision.

Cluster Map of Margins in Singapore's 2020 General Election



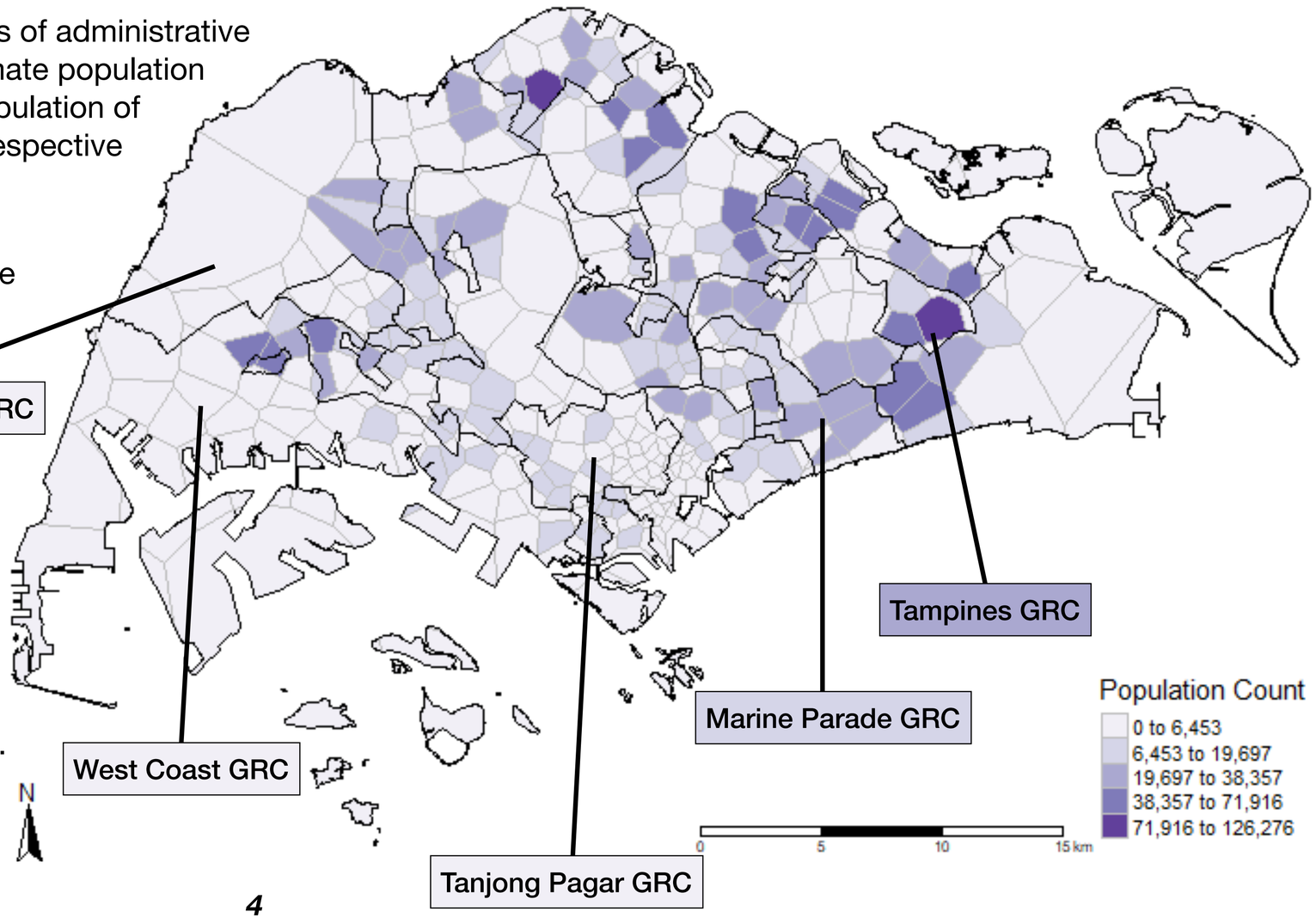
Interpolating Census Data

A Thiessen polygon map was created using centroids of administrative subzones and their associated constituency, to estimate population Characteristics of the respective electorates. The population of each polygon was taken as a proportion of its area respective to its subzone.

Eastern constituencies, such as Tampines and Marine Parade have a higher population density compared with the west, such as Chua Chu Kang and West Coast, as well as central Tanjong Pagar.

This reflects the differences in land-use: Tanjong Pagar contains the shopping district, the west contains military camps, green space, industrial space, while the east has many high-rise housing developments (URA, 2024). This should be considered when analysing vote shares, as the size of the electorate is not proportional to landmass.

Thiessen Polygon Map of Singapore's Administrative and Electoral Boundaries

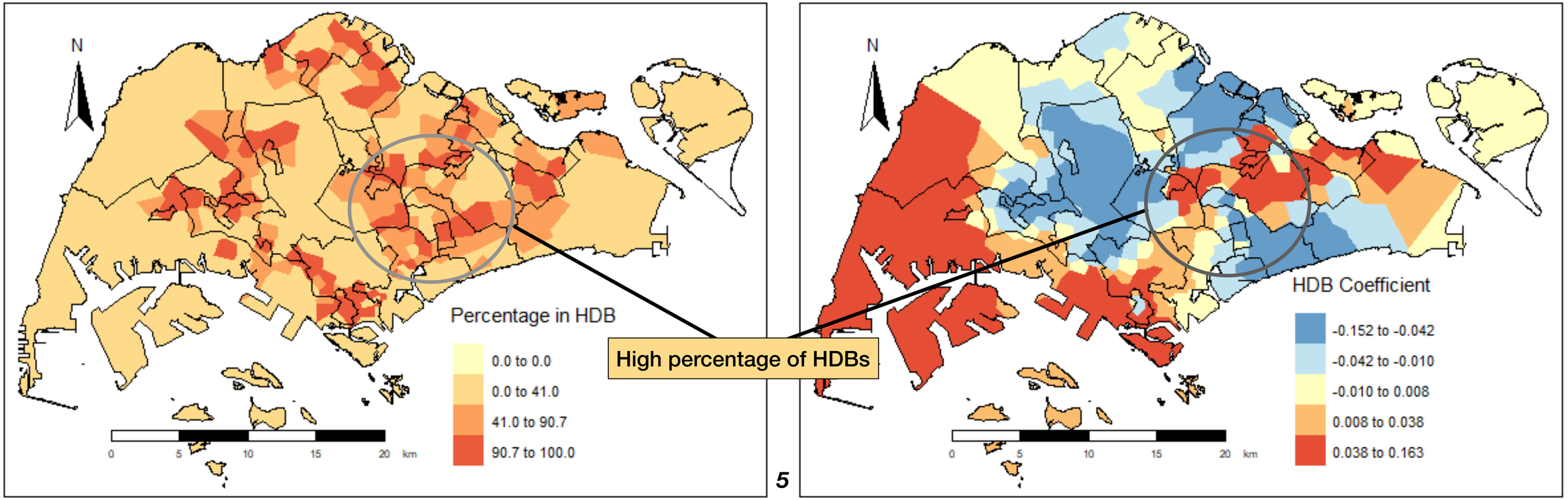


Impact of Housing on Margins

Multicollinearity exists between the proportion of public housing (HDB) and elderly residents, hence separate Geographically Weighted Regressions (GWR) were conducted. GWR captures local regression coefficients between the percentage living in HDB and PAP-leaning voting margins. Constituencies' different sizes was accounted for using an An adaptive kernel bandwidth (Brunsdon *et al*, 1996).

A high percentage of HDB residents highlighted saw coefficients differing in direction across geographies, however their magnitudes were less than one, suggesting no correlation between the proportion of HDB residents and PAP support.

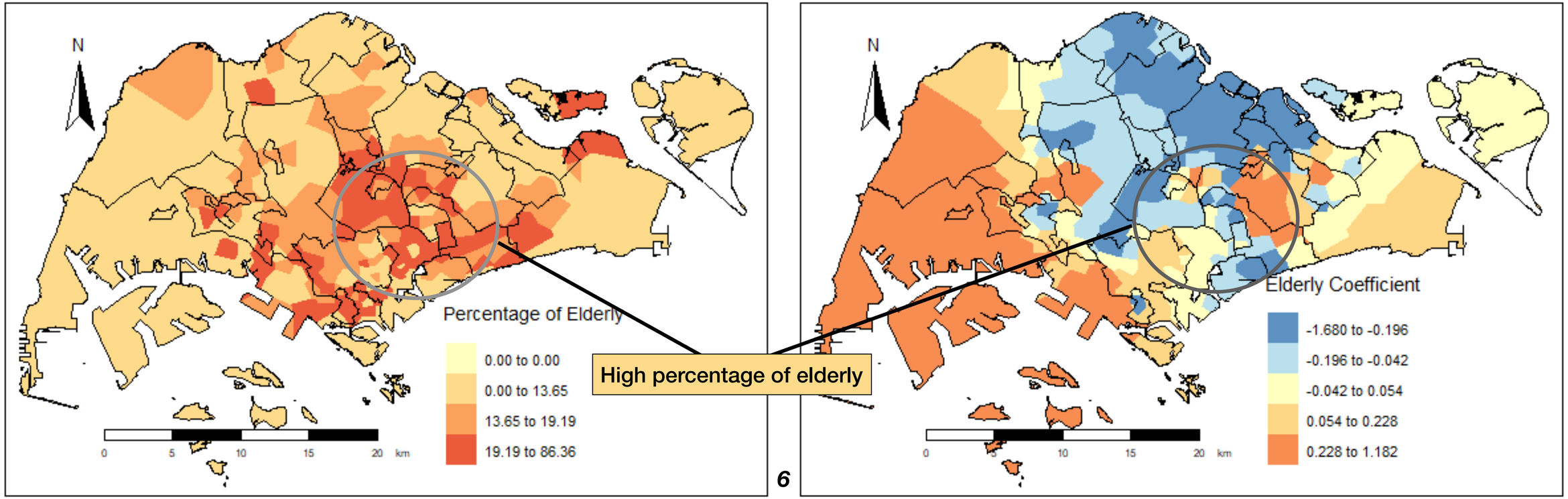
Percentage of HDB population (%) and its Regression Coefficient with PAP-leaning Margins



Impact of Elderly on Margins

As a young country, those aged above 65 have experienced Singapore's entire political development. Theories propose the elderly are loyal to the PAP, due having experienced Singapore's transformation firsthand. Younger citizens, however, have expressed uncertainty about current policies and politics (George & Low, 2020). The relationship between elderly residents and PAP support was analysed with GWR. The correlation has a similar geographic distribution to that of the HDB analysis, however with a greater magnitude. A negative correlation in the northeast indicates the proportion of elderly population is a strong predictor for a WP-leaning margin. Nonetheless, the global correlation coefficient of 0.12 suggests a negligible positive correlation between the elderly demographic and PAP support.

Percentage of Elderly population (%) and its Regression Coefficient with PAP-leaning Margins



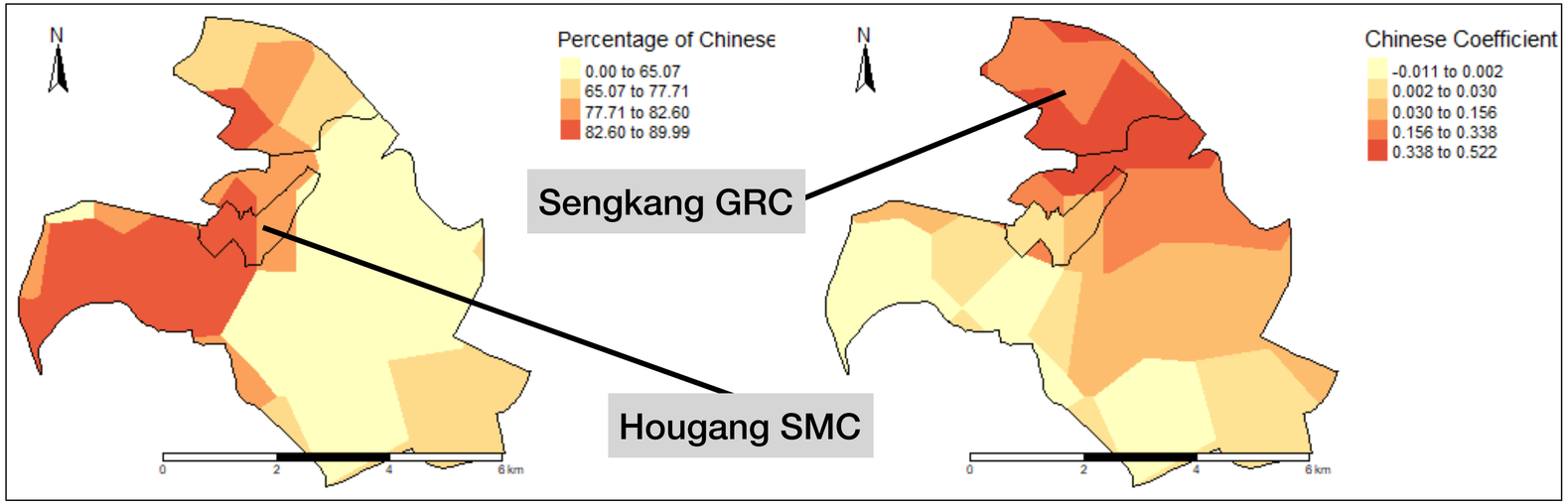
WP Constituencies

It is understood that WP leveraged on the support of elderly, Teochew-Chinese population in the region (Tham, 2020). A combined GWR captures relationship between the proportions of Chinese and above 65 populations, with winning margins.

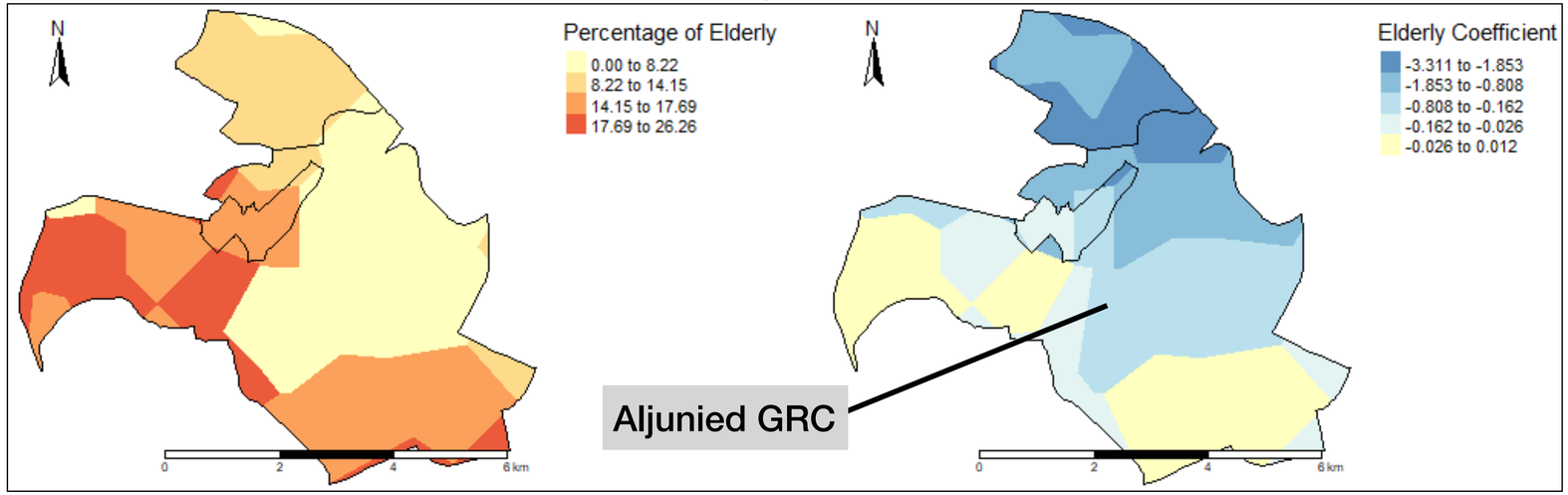
A weak positive relationship exists between the Chinese population and PAP-leaning margins, disagreeing with the hypothesis. Conversely, a negative correlation exists between the elderly population and PAP-leaning margins. Its distribution pattern is similar to that of the Chinese coefficient, but instead in the opposite direction, supporting the hypothesis. Both correlations were most intense in Sengkang, WP's newest ward.

The Chinese coefficient of 0.2 was negligible, suggesting that not all areas exhibit the positive correlation with winning margins. The -0.99 regression coefficient for elderly indicates a weak negative relationship with PAP support.

Chinese Population (%) and Regression Coefficient in WP Wards



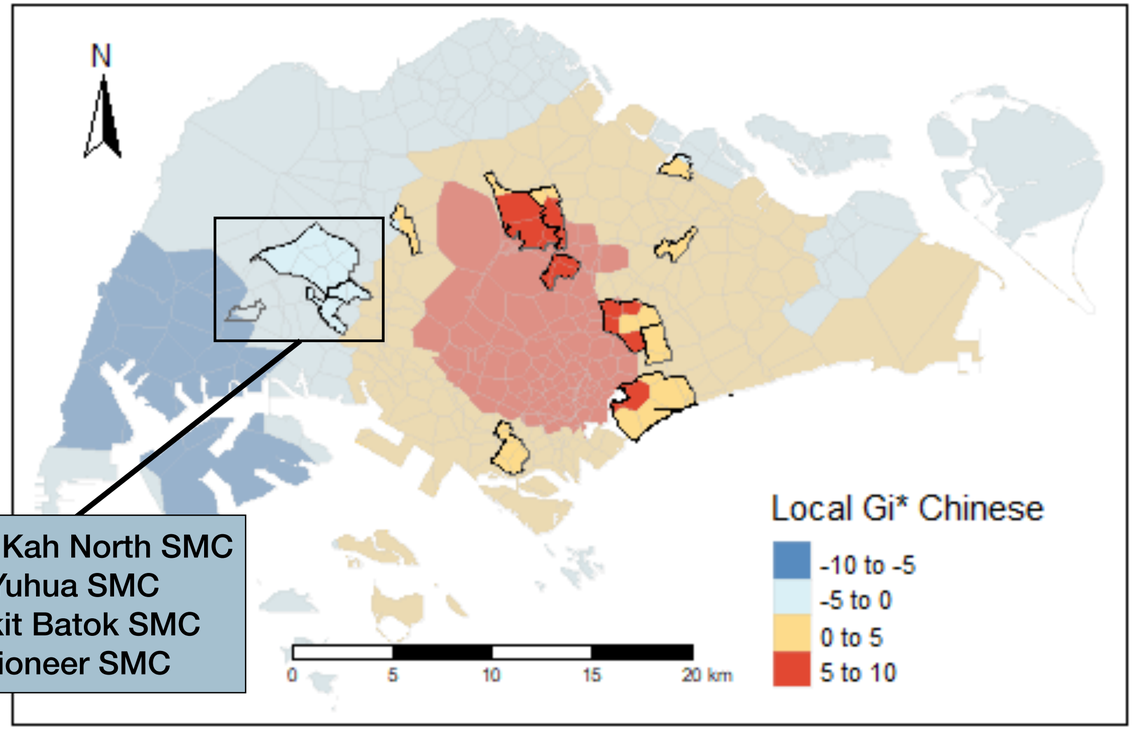
Elderly Population (%) and Regression Coefficient in WP Wards



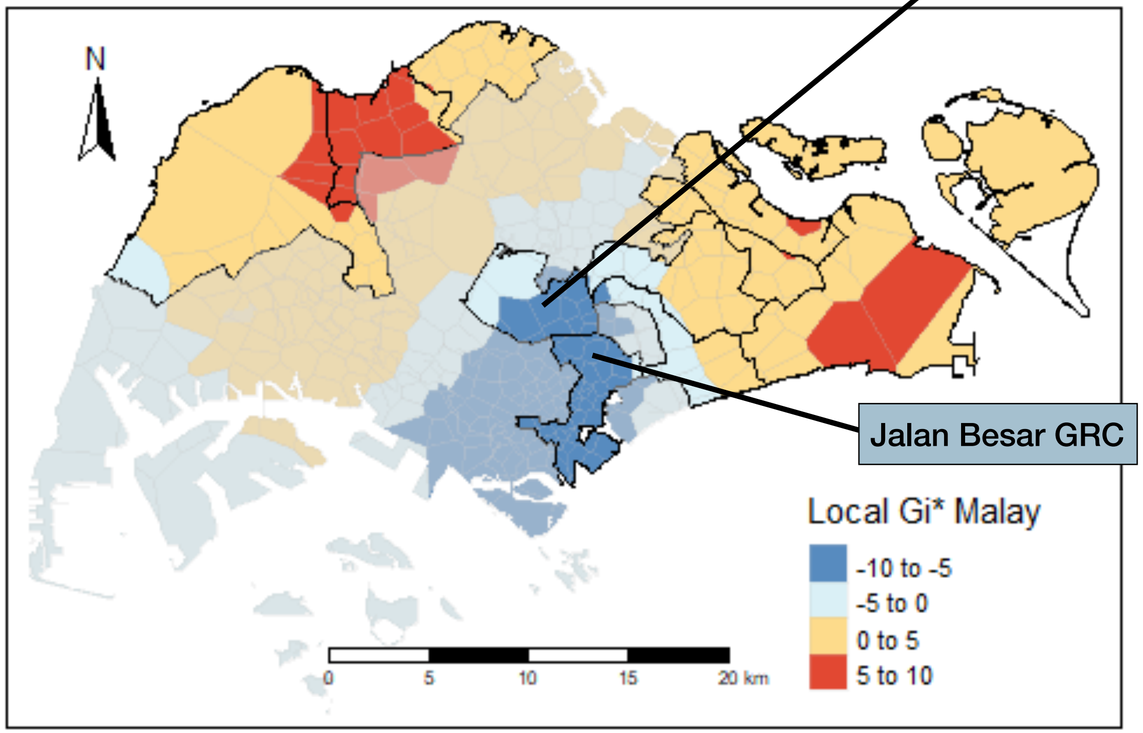
Racial (Mis)representation *Please view together with page 9

To investigate whether the mandated minority representations in GRCs genuinely meet demographic compositions, a local Gertis-Ord was ran for each ethnic group. The subsequent Gi* statistic, represented as a Z-score, identifies statistically significant hot- and coldspots, with a fixed distance threshold of 8km. A positive Gi* indicates spatial clustering, or a high percentage of the respective racial group (hot spots), while a negative score suggests low percentages of a racial group (cold spots). The map displays hot and cold spots of the Chinese, Malay, Indian and Other populations, overlaid with GRCs allocated to their respective race. SMCs were highlighted for the Chinese population.

Hot/Cold Spot Map of Racial Compositions in Singapore with Electoral Boundaries

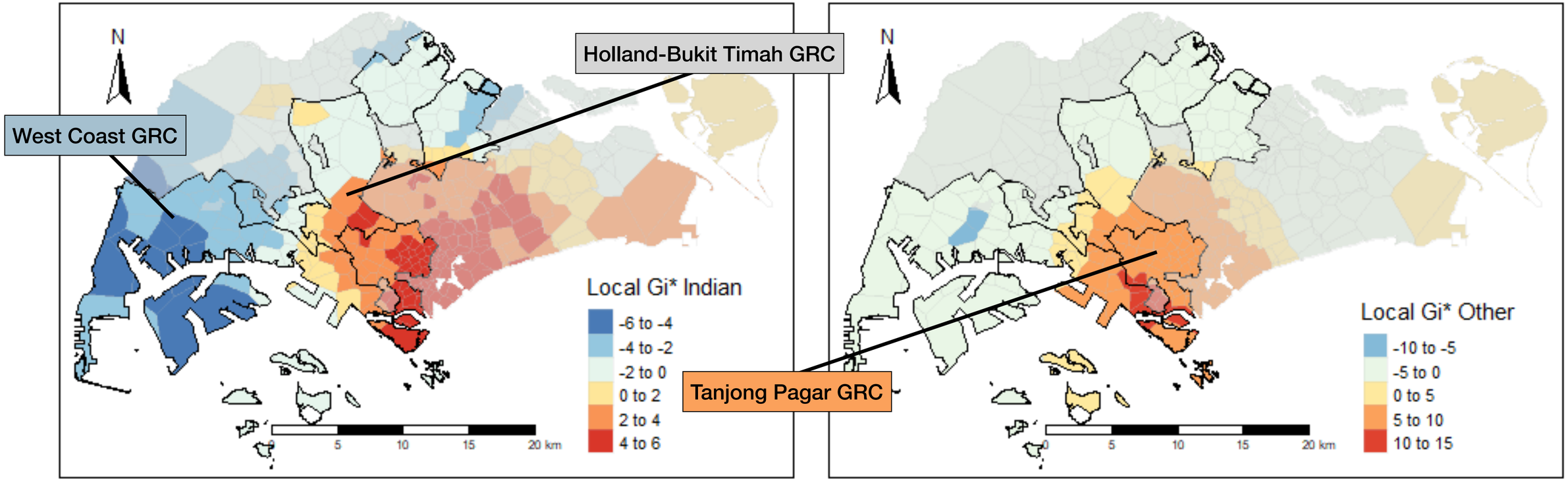


Hong Kah North SMC
Yuhua SMC
Bukit Batok SMC
Pioneer SMC



Bishan-Toa Payoh GRC

Jalan Besar GRC



The Chinese population is concentrated in the heartlands, Malays in the east and northwest, Indians in the southeast, and Others heavily concentrated in the south. The low GI* in the far west across all races supports the low population density and non-residential land-use of the region. While some constituencies' minority threshold were identified as appropriately, others are completely fail to ensure minority representation.

Malay Underrepresentation

The are 4 SMCs identified are not only cold spots for the Chinese population, but is also a Malay cluster. Furthermore, Malay GRCs Bishan-Toa Payoh and Jalan Besar are Malay cold-spots, and Chinese and Indian hot-spots respectively. Part of the northwestern Malay hot-spot falls in Holland-Bukit Timah, an Indian/Other GRC. Holland-Bukit Timah has a clear north-south divide between the Malay and Indian/Other communities, respectively. Similarly, an east-west divide is seen in West Coast.

The spatial mismatch of minority wards and distribution of communities uncover inequitable representation of the minority population, particularly Malays, pointing towards the presences of gerrymandering.

Conclusion

This Atlas revealed the distribution of voter behaviour and demographic dynamics during Singapore's 2020 General Election. LMI revealed PAP's strongholds and its potential threats, however more granular scale analysis in needed. The GWRs elucidated geographic patterns of HDB residents and the elderly population on voting margins, where little correlation between variables were found. The WP-focused GWR revealed a negative correlation between the elderly and PAP support in Sengkang, and no correlation between HDB residents and PAP support. Moreover, scrutiny of minority representation in GRCs reveals race-related inequities, particularly regarding underrepresentation of the Malay community. This raises concerns about the government's motivations behind shifting boundaries, unveiling signs of gerrymandering.

These findings underscore the importance of assessing geographic and demographic characteristics to ensure fair and democratic elections. As Singapore prepares for its next election, greater transparency is needed to help the electorate discern shifting voter sentiments, identify potential opposition strongholds and assess the government's strategies in strengthening its mandate.

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